SESSION 1 • 12 MAY

Program Orientation

& AI Landscape Overview

5-Week Intensive AI Engineering Bootcamp

Weeks 1–5 • 16 Sessions • 5 Expert Mentors

What We'll Cover Today

01

Program Overview & Structure

5 weeks, 16 sessions, how it all fits together

02

The AI Landscape in 2025

Where we are, where we're heading, what matters

03

Your Learning Path

From LLM basics → Production-ready AI systems

04

Meet Your Mentors

Arman, Adkham, Ismail, Daz, Sergey

05

Tools & Setup

What you'll need, pro tips on dev environment

Your 5-Week Journey

Week 1	Week 2	Week 3	Week 4	Week 5
S1–S3	S4–S6	S7–S9	S10–S12	S13–S16
Foundations	**Search & Agents**	**Integrations & Data**	**ML & Deployment**	**Production & Final**
AI Landscape · LLM Basics · Architecture	RAG · Vector DBs · LangChain · LlamaIndex	APIs · Multi-Agent · Data Pipelines	ML Paradigms · GenAI Models · CI/CD & Docker	Monitoring · Multimodal AI · Final Projects

💡 *Each week builds on the last — from theory to shipping real AI apps*

Week 1: Foundations

12 – 16 May • 3 Sessions

S1

Tue 12 May

Program Orientation & AI Landscape Overview

Mentor: Arman Grigoryan

S2

Thu 14 May

LLM Fundamentals: Tokens, Context, Temperature & Prompting

Mentor: Adkham Zokhirov

S3

Sat 16 May

Building Production-Ready AI Apps: Architecture Patterns

Mentor: Ismail Salikhodjaev

Sessions 4 – 12: Building Up

Week 2

S4 Semantic Search, Vector Embeddings & RAG Systems

↳ *Arman*

S5 AI Agents: Tools, Memory & Planning Patterns

↳ *Adkham*

S6 Orchestration Frameworks: LangChain / LlamaIndex

↳ *Ismail*

Week 3

S7 External Integrations: APIs, Databases & Webhooks

↳ *Daz*

S8 Multi-Agent Systems, Coordination Patterns & AI Security

↳ *Adkham*

S9 Data Engineering: Scalable Pipelines & Feature Systems

↳ *Adham*

Week 4

S10 ML Paradigms: Supervised, Unsupervised, RL & Self-Supervised

↳ *Daz*

S11 Generative Models: Diffusion, VAEs & GANs Overview

↳ *Arman*

S12 Deployment Pipelines: Docker, CI/CD & Cloud Environments

↳ *Daz*

Week 5: Production & Launch

9 – 14 June • 4 Sessions

S13 Tue 9 Jun

Production ML: Monitoring, Evaluation & Optimization

Mentor: Daz Akbarov

S14 Thu 11 Jun

Multimodal AI

Mentor: Arman Grigoryan

S15 Sat 13 Jun

Final Project Q&A

Mentor: Ismail Salikhodjaev

S16 Sun 14 Jun

Final Project Q&A

Mentor: Ismail Salikhodjaev

🎯 **By the end, you'll have shipped a real AI product — not just toy demos.**

What You'll Actually Learn

A quick mental model for each major topic

🧠 LLMs

Large Language Models

The engine under everything. Understand tokens, context windows, temperature — so you know why models behave the way they do, not just how to call an API.

🔍 RAG

Retrieval-Augmented Generation

Models don't know your data. RAG connects them to it. Give a model access to your docs, database, or product catalog — and get accurate, grounded answers.

🤖 Agents

AI Agents

Instead of answering one question, an agent takes actions: searches the web, calls APIs, writes files, loops until a goal is met. Think of it as giving the model hands.

📊 Data Eng

Data Engineering

AI is only as good as its data. You'll learn how to build pipelines that clean, transform, and serve data at scale — the unglamorous skill that makes everything else work.

🚀 Deployment

Deployment & MLOps

Getting a model to work locally is step 1. Serving it to real users — with monitoring, CI/CD, and rollback — is step 2. Docker and cloud make this repeatable.

🖼️ Multimodal

Multimodal AI

Models that understand images, audio, and text together. GPT-4o, Gemini, Claude — these aren't just text boxes anymore. You'll build apps that see and hear.

Fun Facts & Things Worth Knowing 🎯

Facts, observations, and opinions from the field

📏 Context window = short-term memory

GPT-4 launched with 8K tokens (~6K words). Claude 3.5 supports 200K. A bigger window = more docs, more history, better answers. But bigger ≠ always better — you still pay per token.

🚀 RAG is a real startup opportunity

Every enterprise has 10 years of docs no one can find. A well-tuned RAG system on internal knowledge = genuine business value. This is one of the most fundable AI ideas in 2025.

🎛️ Temperature: 0 = accountant, 1.5 = jazz musician

Low temp = precise, repetitive. High temp = creative, unpredictable. Most production apps run 0.2–0.7. You'll feel the difference the moment you start experimenting.

⚡ 90% of 'AI failures' are prompt failures

The model is usually fine. The prompt is usually bad. This is why prompting gets a full session — it's the highest-ROI skill in the course and the most underestimated.

🐳 Docker will save your life

'It works on my machine' is not a product. Docker packages your app + its environment together. Learn it once properly and you'll never fight with dependency hell again.

📈 AI code tools genuinely increase output

A 2023 GitHub study found developers using Copilot completed tasks 55% faster. The debate isn't whether to use AI tooling — it's which one fits your workflow and hardware.

Tools You'll Use

Must-Haves

- ✓ Python 3.11+ (don't skip the upgrade)
- ✓ VS Code (+ any AI coding assistant you like)
- ✓ Git & GitHub (commit early, commit often)
- ✓ Docker Desktop (you'll thank Week 4 for this)
- ✓ OpenAI / Anthropic API key (get it today)
- ✓ Jupyter Notebooks (for exploration)

⚡ Pro Setup Tips

Use .env files

Never hardcode API keys. python-dotenv is your friend.

WSL2 on Windows

If you're on Windows, WSL2 + VS Code Remote makes life 10x better.

Virtual envs every time

conda or venv per project. Global installs are chaos.

AI coding assistants

Pick what works for your hardware & workflow — Copilot, Cursor, Codeium. Try a few, commit to one.

Your Mentors

🧠

Arman Grigoryan

S1, S4, S11, S14

🤖

Adkham Zokhirov

S2, S5, S8, S9

⚙️

Ismail Salikhodjaev

S3, S6, S15, S16

🚀

Daz Akbarov

S7, S10, S12, S13

All mentors are practitioners — currently building AI products, not just teaching theory.

💬 Ask questions in class, after sessions, or via the group chat

★ MEET THE INSTRUCTOR

FEATURED MENTOR

Arman Grigoryan

CURRENT ROLE

Senior ML Engineer @ EPAM Systems

Lecturer · ML Engineering, IT Park University (Master's)

Konstanz University, Germany

Oxford ML Summer School

🚀 Startup Founder ·
DeepRead AI - Chrome Extension
Yild Productivity app- iOS App

FIND ME

linkedin.com/in/arman-grigoryan-131201173

The AI Landscape, Honestly

What's hype vs. what's real — as of mid-2025

✅ Actually Impressive Right Now

→ Code generation (GitHub Copilot study: 55% faster task completion)

→ Document Q&A and knowledge retrieval (RAG is production-ready)

→ Multimodal: image → text, audio → text

→ Summarization, classification, extraction at scale

→ Semantic search far outperforming keyword search

⚠️ Still Tricky / Overhyped

→ Reliable long-horizon reasoning (still hallucinates confidently)

→ Truly autonomous agents (they drift; need careful guardrails)

→ Real-time data without tools (knowledge cutoff is real)

→ Privacy in production (data leakage risks are non-trivial)

→ Replacing domain experts wholesale (augment, don't replace)

S1 · Today's Session

Program Orientation & AI Landscape Overview

🎯 **Goal**

Understand the AI ecosystem, the current state of LLMs, and career opportunities in AI engineering

📡 **Format**

- Live lecture + Q&A
- Interactive — ask questions any time

🛠️ **Hands-On Task**

- Set up your dev environment
- Run your first LLM API call

🧭 **What You'll Learn**

- The AI ecosystem mapped out
- Current state of LLMs (GPT-4o, Claude, Gemini)
- Career paths & opportunities in AI engineering

Before next session: Get your API key · Set up Python 3.11+ · Install VS Code

Let's Build Something Real.

5 weeks from now you'll understand how ChatGPT works under the hood, be able to build your own RAG pipeline, deploy a model to production, and have a portfolio project that actually impresses hiring managers.

Before next session:

📌 Set up your API keys tonight

💬 Join the group chat

❓ Come with questions to every session

📓 Keep a learning journal — compound interest applies to knowledge too